Lunar Smart Adaptive Habitat Design (NASA HUNCH) — annotated copy

This is a copy of the NASA HUNCH project page. Under every page: which HUNCH requirement it is, what we built for it and how, with pictures from both simulations.

Physical parts (2 ft × 2 ft scale model, foam board, sensor electronics) are built outside the games and are marked as such.



Lunar Smart Adaptive Habitat Design with AI Psychological Monitoring for Crew Adaptive Positive Mental Health

2025-26

BIOMEDICAL
And / Or
Software Projects or Both

■ DONE ✓ WHAT WE DID WELL — Lunar Smart Adaptive Habitat Design (NASA HUNCH 2025-26)

- Built as two working Unity/VR simulators plus the physical model.

Pictures from the simulations:



Realistic game



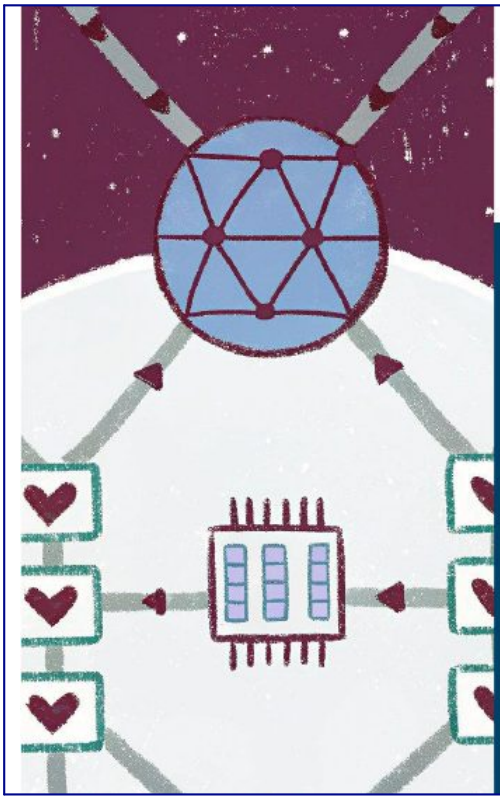
■ DONE ✓ WHAT WE DID WELL — Biomedical and/or software — both

- Software: the ARES AI engine and two games. Biomedical: simulated wearable data (heart rate, HRV, sleep, skin response, cortisol) per crew member.

Pictures from the simulations:



Realistic game



Project Overview

Mission:

Support astronaut **mental health** on the Moon NASA HUNCH empowers students to solve real NASA challenges. This project combines software biomedical (health, well-being) and (AI, monitoring) and approaches to design a next-generation lunar habitat.

■ DONE ✓ WHAT WE DID WELL — Project overview — goal and mission

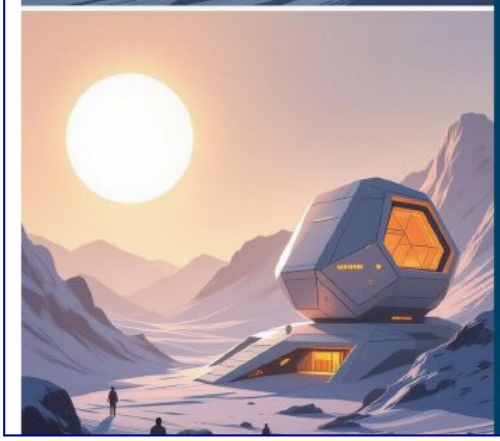
- Psychological research + AI to monitor and support mental health, joined with smart building systems (ECLSS, power, environment) in the same simulation.

Pictures from the simulations:



Realistic game





The Problem

Long lunar missions = isolation, stress, and risk to mental health Current habitats: focus on physical safety, not psychological needs Need: Adaptive, supportive environments for crew well-being Astronauts face unique psychological challenges: isolation, confinement, and stress. Traditional habitats protect physically, but we must also address mental health for mission success.

■ DONE ✓ WHAT WE DID WELL — The problem — isolation, stress, confinement

- Every zone and AI action targets one of the six risk conditions.



The Challenge

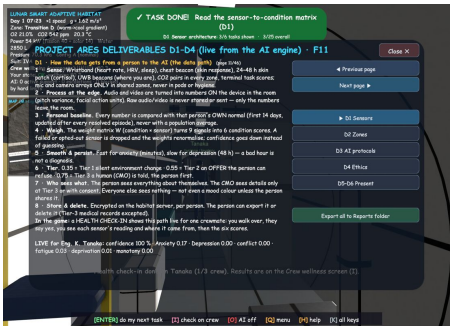
Design a lunar habitat that: Monitors crew psychological health with AI Adapts environment to individual needs Integrates smart building systems (ECLSS, power, environment) Is feasible for lunar deployment Our challenge: blend AI, biomedical monitoring, and smart building tech into a habitat that senses, predicts, and responds to each astronaut's needs—while working reliably on the Moon.

Provide an actual working demo of Personal Space and some Common Spaces for

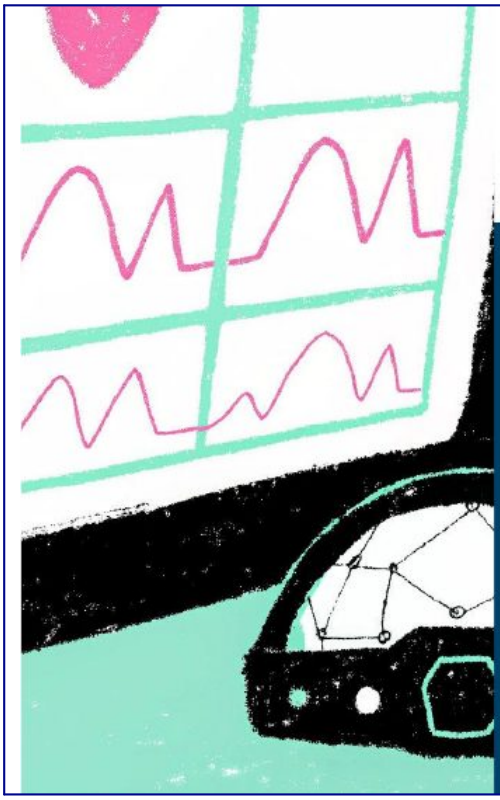
■ DONE ✓ WHAT WE DID WELL — The challenge — monitor, adapt, integrate, feasible

- Monitor: live AI dashboard. Adapt: per-person environment changes. Integrate: ECLSS (CO2 scrubber, water recycler), power and alarms in the same game. Feasible: every sensor has mass and power.

Pictures from the simulations:



Realistic game



Deliverables - Creative Problem Solving

- AI psychological monitoring dashboard
- Adaptive feedback/intervention system
- Biomedical Simulation: Simulated wearable data (heart rate, sleep, mood)
- Personalized countermeasures
- Scale Model 3D Printed habitat, Foam Board Habitat and electronic sensor systems with real time monitoring and reporting.
- Students will deliver a working prototype, a biomedical simulation, or a physical or virtual model.

■ DONE ✓ WHAT WE DID WELL — Deliverables — dashboard, intervention system, simulated wearable data, countermeasures

- AI psychological monitoring dashboard: YES (F1 / Project ARES dashboard).
- Adaptive feedback/intervention system: YES (three tiers, offers, crew tasks).
- Biomedical simulation with simulated wearable data: YES (health check-in shows heart rate, HRV, sleep, skin response, cortisol).
- Personalized countermeasures: YES ('Help' picks what fits that person's highest condition).
- Working demo of personal and common spaces: YES (private sleep dome + communal galley).

Gap / still to do: Scale model / foam board / sensor electronics are physical — done outside the game (Farhan's model).

Pictures from the simulations:



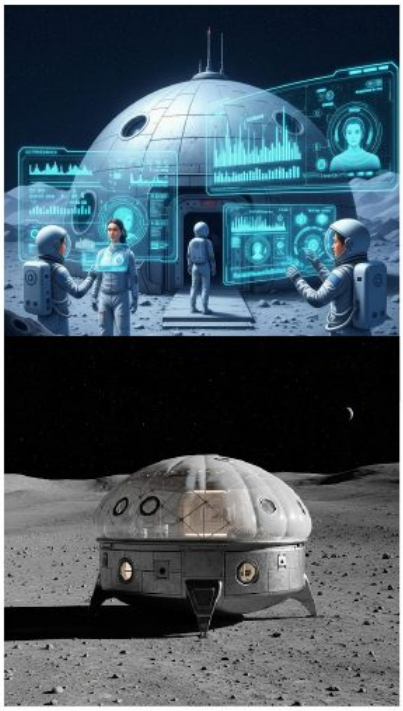
Realistic game



Realistic game



Realistic game: personalised daily wellness tasks



So What would I do First?

- Research NASA Pill Habitat Design for the moon

Feature	Details (Based on NASA Artemis/Surface Habitat Concepts)
Crew Capacity	2 astronauts (standard), expandable to 4 (peak)
Uncrewed Duration	Up to 6–24 months of autonomous operation
Structure	Multi-level: rigid metallic base + inflatable upper deck
Internal Layout	Sleeping, food prep, hygiene, work, storage
Life Support	Advanced ECLSS, waste management, radiation shielding
Power	Solar arrays + nuclear fission backup
Location	Lunar south pole (e.g., Shackleton Crater)
Autonomy	Remote monitoring, automated maintenance, and mode switching

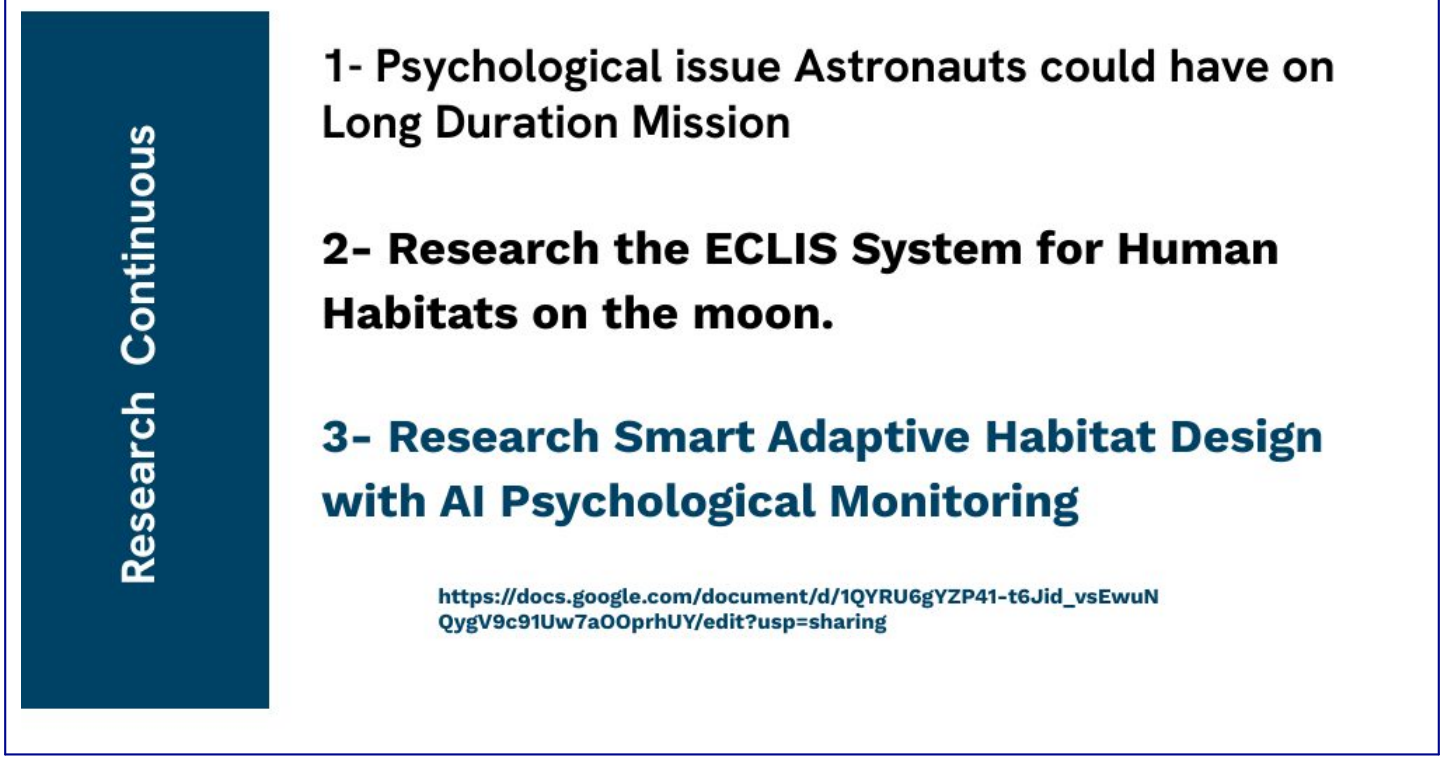
■ DONE ✓ WHAT WE DID WELL — Research NASA pill habitat (2–4 crew, multi-level, ECLSS, south pole)

- Crew of 4 fits the 2–4 capacity; multi-level base with ECLSS, solar + fission power, south-pole setting.

Pictures from the simulations:



Realistic game



■ DONE ✓ WHAT WE DID WELL — Research continuous

- Research PDF (64 pages) drives the numbers used in the games.





Make a Plan and ....

Sketch out a few designs for the outside

Build a Scale Model on a 2ft by 2ft piece of wood

- Foam Board
- Cardboard
- 3D Print

Make sure the roof or floors can be removed to look inside

Make a separate box outside to hold your electronics (Not Part of the Building.

Focus on the Layout

■ DONE ✓ WHAT WE DID WELL — Make a plan — sketch, scale model, removable roof

- The digital twin in Unity lets you look inside every room; the physical scale model is separate.





Make a Plan and ....

Sketch out a few designs for the outside

Build a Scale Model on a 2ft by 2ft piece of wood

- Foam Board
- Cardboard
- 3D Print
- Other

Make sure the roof or floors can be removed to look inside

Make a separate box outside to hold your electronics (Not Part of the Building.

Focus on the Layout

■ DONE ✓ WHAT WE DID WELL — Make a plan (continued)

- Same as above.



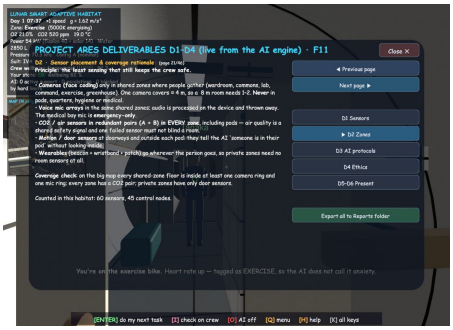
Make your Lunar building …

- Smart
- Adaptive Habitat Design with AI
- With Psychological Monitoring (Voice, Facial, Group Interaction, Lafter, Jokes, tears, isolation)
- Focus on one Room at a time

■ DONE ✓ WHAT WE DID WELL — Smart, adaptive, psychological monitoring (voice, facial, group interaction, isolation)

- Voice prosody and facial coding in shared zones (processed on-device), group interaction and isolation from UWB proximity; room by room design.

Pictures from the simulations:



Realistic game



Key Considerations

Design considerations for a space habitat:

- **Dimensions**: ~20–120 m³ per crew member
- **Structure**: Modular, expandable, radiation-shielded
- **AI Monitoring**: Real-time, privacy-respecting, adaptive
- **Smart Systems**: ECLSS, environmental, and power controls
- **Psychological Support**: VR/AR, adaptive lighting, social prompts

Must provide a Physical Design demonstrating Individual Space and Some Common Space with the application of destressors and feedback.

Design must balance comfort, safety, and efficiency. AI must be ethical and respect privacy. Smart systems should be robust and easy to use, supporting both physical and mental health.

More Info at:
https://docs.google.com/document/d/1QYRU6gYZP41-t6Jid_vsEwuNQygV9c91Uw7aOOprhUY/edit?usp=sharing

■ DONE ✓ WHAT WE DID WELL — Key considerations — 20–120 m³ per crew, modular, privacy-respecting AI, VR/AR

- Privacy-respecting AI (no sensors in pods, opt-out, export, delete), VR support (Meta Quest), adaptive lighting and social prompts.

Pictures from the simulations:



Realistic game

Constraints & Hints





Lunar Environment Constraints

The **lunar environment** presents several constraints: **extreme temperatures**, **radiation**, pervasive **dust**, and **limited resupply** options from Earth. Designs must be resilient and resourceful to overcome these challenges, focusing on **autonomy** with minimal Earth intervention and efficient use of **lunar materials** and power. **Scalability** is essential for accommodating larger crews.

Design Hints

Leverage research from **ISS/Artemis habitats** and utilize **open-source tools** like **Unity** and **Arduino** to expedite development. Prioritize **user experience** to ensure designs are astronaut-friendly. The **Moon is harsh**, so designs must be **modular**, **autonomous**, and **efficient**. Focus on available technology to enhance the astronaut experience while ensuring resilience.

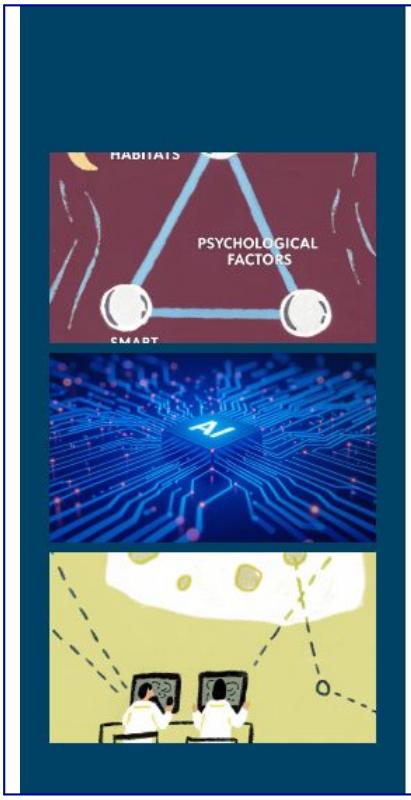
■ DONE ✓ WHAT WE DID WELL — Constraints & hints — Unity, astronaut-friendly

- Built in Unity as suggested. Made astronaut-friendly tonight: one key (ENTER / G) does the next task, simple 7-button menu, short key bar.

Pictures from the simulations:



Realistic game: simple menu



Example Project Flow

Research

Conduct thorough **research** on **psychological factors**, **lunar habitats**, and **smart systems** to lay a solid foundation for the project.

More Info at:
https://docs.google.com/document/d/1QYRU6gYZP41-t6Jid_vsEwuNQygV9c91Uw7aOOprhUY/edit?usp=sharing

Design

Create a detailed **design** by sketching the **layout**, defining **AI parameters**, and planning **integration** to guide the development process.

Prototype, Test, Present

Develop a **model/VR** prototype, **develop software**, **simulate scenarios**, **refine interventions**, and then **present** a **demo**, **walkthrough**, and **report**.

■ DONE ✓ WHAT WE DID WELL — Example project flow — research, design, prototype/test/present

- Prototype = two simulators; test = automated playtests of every mission and button; present = the review-board run-of-show built into the game.

Pictures from the simulations:



Realistic game

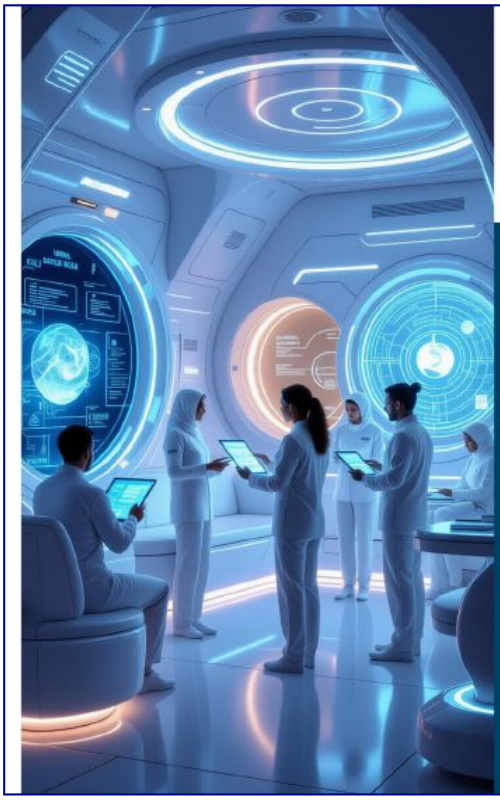


Conclusion

Students innovate for real NASA needs Combining, biomedical, software and engineering skills Supports future lunar explorers' mental and physical health This project is a chance to make a real impact on the future of space exploration. By supporting astronaut well-being, you're helping pave the way for humanity's next giant leap.

■ DONE ✓ WHAT WE DID WELL — Conclusion

- Biomedical + software combined in one working demo.



Questions & Discussion

Ready to launch your ideas?

Let’s discuss your concepts and questions!!

More Info at:

https://docs.google.com/document/d/1QYRU6gYZP41-t6Jid_vsEwuNQygV9c91Uw7aOOprhUY/edit?usp=present

■ DONE ✓ WHAT WE DID WELL — Questions & discussion

- Board questions Q1–Q6 rehearsed live.

Pictures from the simulations:



Realistic game

Other Details

First Floor

- Airlock/Entry
- Storage
- Laboratory/Workspace
- Hygiene Area

Second Floor

- Crew Quarters (individual sleeping pods)
- Galley/Wardroom (dining and social area)
- Additional Storage

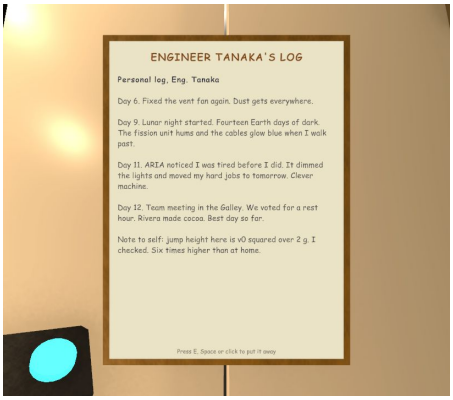
The spatial organization separates noisy (work/lab) and quiet (sleeping) areas, with hygiene and galley areas placed for easy access.



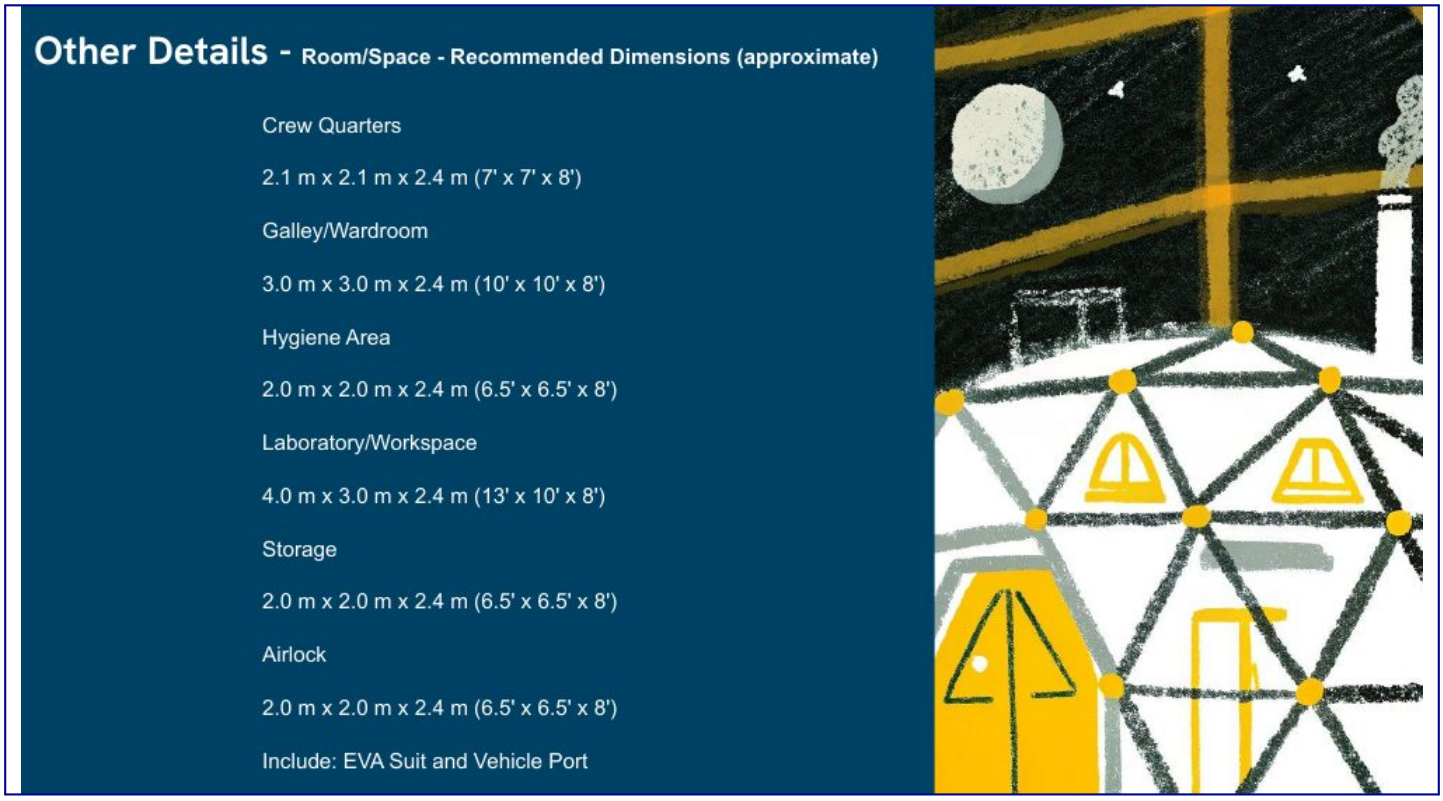
■ DONE ✓ WHAT WE DID WELL — Floor organisation — quiet vs noisy areas

- Sleeping quarters are separated from the work/lab areas; hygiene and galley are easy to reach.

Pictures from the simulations:



Sci-fi game: crew quarters (central dorm kept)



■ PARTLY — see gap WHAT WE DID WELL — Recommended room dimensions

- Rooms in both bases follow these sizes closely (quarters about 2.1 m pods, galley about 3 m, lab about 4 × 3 m, airlock with EVA suit port).

Gap / still to do: Put the exact measured dimensions on the printed floor plans.

Research

Key for this project is to start to Research in:

- **Psychological Support**: VR/AR, adaptive lighting, social prompts
- **AI Monitoring**: Real-time, privacy-respecting, adaptive, Psychological and Physical
- **Smart Systems**: ECLSS, environmental, and power controls
- **Structure**: Modular, expandable, radiation-shielded



■ DONE ✓ WHAT WE DID WELL — Research areas — psychological support, AI, smart systems, structure

- All four areas are in the games: VR nature and adaptive lighting, real-time privacy-respecting AI, ECLSS/power controls, modular base.